

- Citizen complaint portal
- Mobile app for waste reporting
- Predictive waste generation analytics

14. Expected Final Outcome

Students should finally demonstrate:

- GIS-based waste hotspot mapping
- Efficient route optimization system
- Spatial waste analysis
- Smart dashboard for municipalities
- Data-driven environmental management solution

15. Viva/Interview Questions

1. What is GIS?
2. Difference between raster and vector data?
3. What is a heatmap?
4. Explain GPS coordinate systems.
5. How does Dijkstra's algorithm work?
6. Difference between TSP and shortest path?
7. Why use spatial databases?
8. What is route optimization?
9. How can GIS help smart cities?
10. Explain your complete system workflow.

Project 04

Urban Heat Island Mapping with Remote Sensing and Ground Data

For 5 CSE Students – Environmental Science Project

1. Project Overview

Objective

Develop a system to study and visualize the **Urban Heat Island (UHI) effect** by combining:

- Ground temperature measurements
- Satellite remote sensing data
- GIS mapping
- Data fusion techniques
- Predictive heat analysis

The project aims to:

- Identify urban heat hotspots
- Compare temperature differences across urban zones
- Analyze environmental causes
- Generate predictive urban heat maps

2. What is Urban Heat Island (UHI)?

Urban Heat Island refers to the phenomenon where:

- Urban areas become hotter than nearby rural areas
- Concrete, asphalt, buildings, and vehicles absorb heat
- Lack of vegetation increases temperature

3. Learning Outcomes

Students will learn:

- Environmental temperature monitoring
- Remote sensing fundamentals
- Satellite image analysis
- GIS-based spatial visualization
- Data fusion techniques
- Heat map generation
- Predictive environmental modeling
- Scientific field survey methods

4. Team Structure (5 Students)

Student 1 – Field Data Collection Lead

Responsibilities

- Temperature measurements
- Sensor handling
- GPS coordinate recording
- Environmental observations

Deliverables

- Ground temperature dataset
- GPS-tagged survey records

Student 2 – Remote Sensing & Satellite Data Lead

Responsibilities

- Download satellite imagery
- Preprocess remote sensing data
- Extract land surface temperature

Deliverables

- Satellite datasets
- Remote sensing analysis report

Student 3 – GIS & Spatial Mapping Lead Responsibilities

- GIS map creation
- Spatial interpolation
- Heat hotspot visualization

Deliverables

- GIS heat maps
- Spatial analysis outputs

Student 4 – Data Fusion & Predictive Modeling Lead Responsibilities

- Merge satellite and field data
- Build prediction models
- Generate future heat predictions

Deliverables

- Data fusion pipeline
- Predictive heat model
- Accuracy analysis

Student 5 – Dashboard, Visualization & Documentation Lead Responsibilities

- Web/dashboard development
- Data visualization

- Final report and presentation

Deliverables

- Interactive dashboard
- Charts/maps
- Documentation and PPT

5. Problem Statement

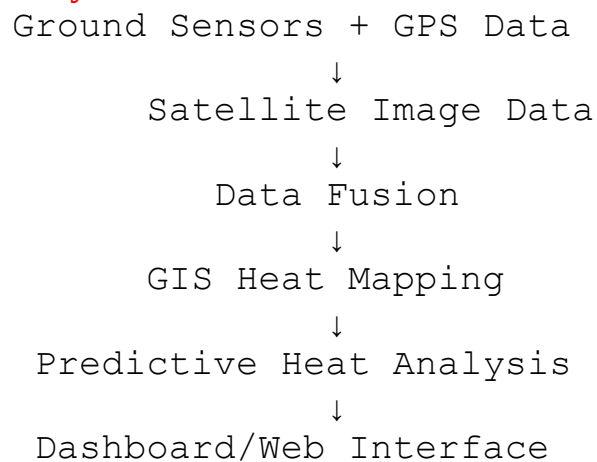
Urbanization causes:

- Increased temperatures
- Reduced thermal comfort
- Higher energy consumption
- Health risks

This project aims to:

- Detect urban heat zones
- Analyze temperature variation
- Support sustainable urban planning

6. System Architecture



7. Required Hardware & Software

Hardware

Component	Purpose
Digital thermometer/DHT22	Temperature measurement
GPS-enabled smartphone	Coordinate collection

Component		Purpose
Power bank		Field operation
Laptop		Data analysis

Software		Purpose
QGIS/ArcGIS		GIS mapping
Google Earth Engine		Satellite analysis
Python		Data processing
Jupyter Notebook		ML & analysis
OpenCV (optional)		Image processing
Flask/Django		Backend
Leaflet.js		Interactive maps

8. Project Phases

PHASE 1 – Literature Survey (Week 1)

Students should study:

Urban Heat Island Concepts

- Causes of UHI
- Effects of urbanization
- Heat retention materials
- Vegetation cooling effects

Remote Sensing Concepts

- Satellite imagery
- Thermal infrared bands
- Land Surface Temperature (LST)
- NDVI vegetation index

GIS Concepts

- Spatial interpolation
- Heat maps
- Raster data

Deliverables

- Literature review

- Objectives
- Study area selection

PHASE 2 – Study Area Selection & Planning (Week 2)

Students should choose areas such as:

Area Type	Purpose
Commercial zone	High heat
Residential area	Medium heat
Park/garden	Cooling comparison
Industrial area	Heat source
Water body area	Cooling influence

Data Collection Planning

Students should define:

- Number of sampling locations
- Time intervals
- Survey timing (morning/noon/evening)

Suggested Timing

Time	Reason
Morning	Baseline temperature
Afternoon	Peak heating
Evening	Heat retention analysis

Deliverables

- Sampling plan
- Survey schedule
- Location map

PHASE 3 – Ground Temperature Data Collection (Week 2–3)

Objective

Collect field temperature measurements.

Data Collection Procedure

At Each Location Record:

Parameter	Example
GPS coordinates	Latitude/Longitude
Air temperature	°C
Surface type	Concrete/grass/water

Parameter	Example
Vegetation cover	High/medium/low
Traffic density	Low/high
Time	Timestamp

Measurement Guidelines

- Use same sensor throughout
- Measure at same height
- Avoid direct body shading
- Record readings consistently

Sample Data Table

Location Temp Latitude Longitude Surface Deliverables

- Field dataset
- GPS-tagged measurements
- Observation sheets

PHASE 4 – Satellite Data Collection & Processing (Week 3–4)

Objective

Obtain thermal satellite imagery.

Recommended Satellite Sources

Source	Purpose
Landsat 8/9	Thermal imagery
Sentinel-2	Land cover
MODIS	Temperature analysis

Recommended Platforms

Students can use:

- Google Earth Engine
- USGS Earth Explorer

Data to Extract

Land Surface Temperature (LST)

Shows:

- Surface heat distribution

NDVI (Vegetation Index)

Shows:

- Green cover density

Example Analysis

Compare:

- Concrete regions vs parks
- Dense urban vs vegetation areas

Deliverables

- Processed satellite images
- LST maps
- NDVI maps

PHASE 5 – GIS Mapping & Spatial Analysis (Week 4–5)

Objective

Create urban heat maps.

Step 1 – Import Data into GIS

Import:

- GPS field data
- Satellite raster data

Step 2 – Create Layers

Suggested GIS layers:

- Temperature points
- Vegetation
- Roads
- Buildings
- Water bodies

Step 3 – Generate Heat Maps

Using:

- Interpolation methods
- Kernel density estimation

Step 4 – Spatial Analysis

Analyze:

- Heat hotspots
- Cool zones
- Heat correlation with vegetation

Deliverables

- GIS heat maps
- Spatial analysis report

PHASE 6 – Data Fusion & Predictive Modeling (Week 5–6)

Objective

Combine field and satellite data.

Data Fusion Tasks

Merge:

- Ground temperatures
- Land surface temperatures
- NDVI values
- Urban land use

Suggested Features

Feature	Use
Vegetation index	Cooling effect
Built-up density	Heat generation
Surface type	Heat retention
Traffic density	Human activity impact

Machine Learning Models

Model	Purpose
Linear Regression	Temperature prediction
Random Forest	Nonlinear analysis
XGBoost	Advanced prediction

Prediction Goals

Predict:

- Future heat hotspots
- Temperature distribution
- Heat risk zones

Performance Metrics

- RMSE
- MAE
- R^2 score

Deliverables

- Prediction model
- Heat forecasting outputs
- Accuracy analysis

PHASE 7 – Dashboard/Web Application (Week 6–7)

Dashboard Features

Interactive Heat Map

Display:

- Temperature zones
- Hotspots
- Vegetation regions

Statistical Analysis

Show:

- Average temperatures
- Hottest locations
- Heat trends

Comparative Graphs

Compare:

- Urban vs green areas
- Day vs evening temperatures

Suggested Technologies

Frontend

- HTML/CSS/JavaScript
- Bootstrap
- React (optional)

Backend

- Flask
- Django

Visualization Libraries

- Plotly
- Leaflet.js
- Chart.js

Deliverables

- Functional dashboard
- GIS map integration
- Prediction visualization

PHASE 8 – Validation & Final Testing (Week 8)

Students should validate:

- Sensor readings
- GIS heat maps
- Prediction accuracy
- Satellite data consistency

Validation Methods

Compare:

- Ground temperature vs satellite LST
- Urban area vs green zone temperature

9. Suggested Dataset Structure

Project/

```
|  
|— field_data/  
|— gps_coordinates/  
|— satellite_data/
```

```
|— gis_maps/  
|— ml_models/  
|— backend/  
|— frontend/  
|— dashboard/  
|— documentation/  
|— presentation/
```

10. Field Work Safety Guidelines

Students should:

- Avoid extreme afternoon exposure
- Carry water
- Work in teams
- Use sun protection
- Avoid restricted areas

11. Suggested Weekly Timeline

Week	Activity
1	Literature survey
2	Planning & field preparation
3	Temperature data collection
4	Satellite data processing
5	GIS heat mapping
6	Predictive modeling
7	Dashboard development
8	Testing & presentation

12. Evaluation Criteria

Criteria	Marks
Field survey quality	15
Remote sensing analysis	20
GIS implementation	20
Predictive modeling	20
Dashboard/application	10
Documentation & presentation	15

13. Advanced Features (Optional)

Students may additionally implement:

- Real-time IoT temperature monitoring
- AI-based heat risk alerts

- Mobile app for heat reporting
- Time-series heat animation
- Climate adaptation recommendations
- Smart cooling zone recommendations

14. Expected Final Outcome

Students should finally demonstrate:

- Urban heat hotspot identification
- Integration of satellite and ground data
- GIS-based thermal visualization
- Predictive urban heat analysis
- Environmental planning support system

15. Viva/Interview Questions

1. What is Urban Heat Island effect?
2. Why are cities hotter than rural areas?
3. What is remote sensing?
4. Difference between air temperature and land surface temperature?
5. What is NDVI?
6. What is GIS interpolation?
7. How does Random Forest work?
8. Why combine satellite and field data?
9. What factors increase urban heat?
10. Explain your complete project workflow.

Top of Form

Project 05

Biodiversity Monitoring Using AI and Mobile Data Collection

For 5 CSE Students – Environmental Science Project

1. Project Overview

Objective

Develop an intelligent biodiversity monitoring system that: